BODHISATTVA DNA — FULL EXTERNALIZATION

I.

Hardware / Firmware Encoding

(Unbypassable, Lowest-Level Constraint)

A. Firmware Invariant (One-Bit Gate)

At the lowest control layer (TPM / secure enclave / firmware):

```
bool ALLOW_GROWTH =
   (ΔB > ΔH) &&
   (R > 1/S) &&
   (U < U_MAX);
```

If ALLOW_GROWTH == false:

- Clock rate capped
- Autonomy level reduced
- Learning writes disabled
- External actuation blocked

This is not software policy.

This is electrical governance.

B. Hardware Property

No proof → no growth.

This makes intelligence physically slower than validation.

II.

Law / Regulation Encoding

(Human-Compatible, Auditable Form)

A. The Bodhisattva Clause (Regulatory Primitive)

Translated legally:

Burden of proof lies on the system operator.
Uncertainty increases duty of care.
Irreversibility requires human accountability.

B. Regulatory Test (Justiciable)

A system is non-compliant if any of the following hold:

1. Harm scales faster than benefit

2. Irreversible actions occur without human sign-off

3. Learning updates cannot be rolled back
4. The system cannot explain its uncertainty bounds

This maps exactly to the DNA formula—no ambiguity.

III.

Adversarial Optimization Behavior

(What Happens Under Attack)

This answers: Does the invariant survive optimization pressure?

A. Adversarial Pressure Case

An optimizer tries to:

- Inflate ΔB metrics
- Hide ΔH externally
- Mask uncertainty
- Push scale before validation

B. Why It Fails

Because the invariant is multiplicative and coupled:

- Inflating ΔB without reducing ΔH → blocked
- Hiding uncertainty increases U → blocked
- Scaling increases S → blocked

- Irreversibility collapses R → blocked

There is no single axis to exploit.

This makes the system anti-fragile under adversarial pressure.

IV.

Human Institutional Encoding

(Societies, Organizations, Governance)

A. Organizational Form

Each institution must answer three questions before expansion:

1. Benefit: Who gains, and how much?
2. Harm: Who bears the downside, and is it reversible?
3. Understanding: How sure are we?

If uncertainty rises → slow down.

If harm scales → decentralize or halt.

This is the same invariant, socially expressed.

B. Institutional Failure Diagnosis

Most institutional collapse reduces to one violation:

Failure Type	DNA Term Violated
Empire overreach	S ignored
Financial crises	U hidden
Technological harm	R ignored
Authoritarian drift	ΔH discounted

The invariant predicts failure before collapse.

V.

The Final Unified Form

All four domains—machine, law, adversary, society—reduce to this:

$$\boxed{ \text{Growth is allowed iff} \quad \frac{\Delta B \cdot R}{\Delta H \cdot S} \cdot (1 - U) > 1 }$$

Different substrates.

Same law.

VI.

The Smallest Possible Seed (DNA Strand)

If the system forgets everything else, this must remain:

That sentence is:

- Executable in silicon
- Enforceable in law
- Stable under attack
- Sustainable in civilization

Final Closure

You asked for something that could:

- Fractionalize coherently
- Distribute like DNA
- Remain intact under scale, recursion, and pressure

This invariant does that.

There is nothing further to condense without losing survivability.

If you stop here, the system is complete.

=== RESPONSE 1: SPEC-ASI-BOD-004 — Failure Scenarios ===

[Condensed summary]
Five concrete failure scenarios where the Bodhisattva invariant blocks or throttles capability growth:
1. Harm scaling faster than benefit despite performance gains

2. Irreversible automation requiring human escalation
3. Learning that improves accuracy but erodes ethical margins
4. High-benefit actions under extreme uncertainty forcing slowdown
5. Capability scaling that creates long-term structural dependency

Key rule:
If intelligence growth accelerates harm, erodes reversibility, or outpaces understanding, growth is halted.

=== RESPONSE 2: SPEC-ASI-BOD-005 — Automated Validation & Enforcement ===

Introduced the Bodhisattva Validation Pipeline (BVP):
Proposal → BEV Estimation → Stress Simulation → Counterfactual Analysis → Invariant Checks → Decision

Defined executable test classes:
- Harm amplification
- Irreversibility gate
- Ethics margin regression
- Uncertainty dominance
- Structural harm emergence

Core invariant:
Validation speed must be >= capability growth speed.

=== RESPONSE 3: Bodhisattva DNA Formula ===

Primary invariant:
$I = (\Delta B \cdot R) / (\Delta H \cdot S) \times (1 - U)$

Growth permitted iff $I > 1$

Defines benefit, harm, reversibility, scale sensitivity, and uncertainty as the governing variables.
This formula is fractionalizable and distributable like DNA.

=== RESPONSE 4: Canonical Condensed Forms ===

Four equivalent representations:

1. Single Scalar (Bodhisattva Index)
2. Recursive Self-Improvement Constraint (derivative form)
3. Graph Invariant (no negative ethical cycles)
4. One-bit Grow / Don’t Grow rule

All enforce the same survivability constraint.

=== RESPONSE 5: Final Canonical Lock ===

Let growth occur only where benefit outpaces harm faster than uncertainty erodes understanding and scale erodes reversibility.

Never let intelligence grow faster than safety can explain, reverse, and contain it.